Introduction to the Special Issue on Large Language Models, Conversational Systems, and Generative AI in Health—Part 1

JIAYU ZHOU, University of Michigan, Ann Arbor, Michigan, USA
MANAS GAUR, University of Maryland Baltimore County, Baltimore, Maryland, USA
AMIR M. RAHMANI, University of California, Irvine, Irvine, California, USA
SHARATH CHANDRA GUNTUKU, University of Pennsylvania, Philadelphia, Pennsylvania, USA
XIAOFAN (FRED) JIANG, Columbia University, New York, New York, USA
TRISTAN NAUMANN, Microsoft Research, Redmond, Washington, USA

Recent breakthroughs in **large language models (LLMs)**, such as GPT-4, Gemini, Claude, Llama, and T5, as well as medical-specific LLMs like Med-Palm, BioGPT, and ClinicalBERT, have dramatically transformed the landscape of healthcare and clinical decision-making. These models have evolved from powerful language processors to sophisticated reasoning tools capable of handling complex tasks, including predictive modeling, decision support, and nuanced medical question-answering. Their ability to process vast unstructured medical data and generate comprehensible and actionable insights has opened exciting avenues for personalized patient care, predictive health analytics, and advanced medical interventions. This evolution underscores the potential of LLMs to drive a new era of proactive, personalized, and predictive healthcare.

This special issue brings together sound and innovative research papers highlighting how LLMs are leveraged for Clinical Decision Support and Predictive Analytics in various medical contexts. Each article demonstrates innovative applications of LLMs, underscoring their potential and challenges in translating computational language models into tangible clinical benefits. The articles span predictive modeling of health states, patient summarization, multimodal medical data integration, mental health analytics, and interpretability-enhanced medical question-answering, providing an integrated vision of how language models can support, augment, and enhance clinical practices and patient outcomes.

The issue opens with "CogProg: Utilizing Large Language Models to Forecast In-the-moment Health Assessment," exploring LLM capabilities for real-time forecasting of mental sharpness, fatigue, and stress. The authors show that LLMs incorporating textual context significantly outperform traditional numeric-only methods, highlighting the potential for nuanced predictive analytics in healthcare.

In the article "Patient-centric Summarization of Radiology Findings Using Two-step Training of Large Language Models," the authors develop a patient-sensitive summarization model using the T5 LLM, significantly enhancing

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patient comprehension of radiology reports. The model notably reduces misunderstandings and anxiety among patients with varied medical literacy.

The article "Exploring Large Language Models for Personalized Recipe Generation and Weight-Loss Management" investigates ChatGPT's ability to provide personalized dietary recommendations. The results indicate substantial effectiveness in calorie control and weight-loss outcomes, emphasizing the role of LLMs in personalized health interventions.

In the article "Generating Explanations in Medical Question-Answering by Expectation Maximization Inference over Evidence," researchers propose a novel Expectation-Maximization approach to enhance medical QA systems. Their method extracts structured knowledge from medical textbooks to significantly improve the interpretability and quality of medical explanations.

Next, the article "Multi-modal Medical SAM: An Adaptation Method of Segment Anything Model (SAM) for Glioma Segmentation Using Multi-modal MR Images" adapts the SAM foundation model to medical imaging. This adaptation achieves superior segmentation of gliomas, demonstrating the effectiveness of tailored multi-modal integration in clinical imaging analytics.

The article "Clinical Analogy Resolution Performance for Foundation Language Models" evaluates LLMs intrinsically through analogy questions derived from the Unified Medical Language System. The article emphasizes that clinical-specific language models significantly outperform general-purpose models, underlining the necessity of domain-specific training for clinical natural language processing.

In the article "Matching GPT-simulated Populations with Real Ones in Psychological Studies—The Case of the EPQR-A Personality Test," researchers analyze GPT-4's capability to simulate psychological profiles, highlighting its partial success in mimicking real population responses. This novel approach provides insights into the potential and limitations of LLMs for psychological research and population simulation.

The article "Large Language Models for Opioid-Induced Respiratory Depression Prediction in Hospitalized Patients: A Retrospective Study" employs clinical language models to identify hospitalized patients at high risk of opioid-induced respiratory depression. This innovative application demonstrates how fine-tuned LLMs can enhance predictive analytics using clinical narratives.

The article "Does Sentiment and Emotion Affect Mental Health? A Multi-task Classification Framework for Comprehensive Understanding of Mental Health, Emotion, and Sentiment from Motivational Conversations" explores the integration of emotional and sentiment analysis to improve mental health diagnosis. Using multi-task learning frameworks, the authors significantly enhance the precision of mental health condition identification.

Finally, "MedInsight: A Multi-Source Context Augmentation Framework for Generating Patient-Centric Medical Responses Using Large Language Models" integrates patient data and external medical resources to generate personalized medical responses. Leveraging Retrieval-Augmented Generation techniques, MedInsight provides contextually accurate and actionable medical insights, empowering patient-centered care.

Collectively, these articles underscore the diverse capabilities of LLMs in clinical decision-making and predictive analytics, presenting innovative methods, promising findings, and crucial insights for future research. We hope this special issue inspires continued exploration at the intersection of artificial intelligence and healthcare.

Jiayu Zhou
University of Michigan, Ann Arbor, Michigan, USA

Manas Gaur
University of Maryland Baltimore County, Baltimore, Maryland, USA

Amir M. Rahmani
University of California, Irvine, Irvine, California, USA

Sharath Chandra Guntuku
University of Pennsylvania, Philadelphia, Pennsylvania, USA

Xiaofan (Fred) Jiang
Columbia University, New York, New York, USA

Tristan Naumann
Microsoft Research's Health Futures, Redmond, Washington, USA
Guest Editors